

Regime Viability on the Northern Cod Stock: An Exact Certification of Obstruction Structure on Public Assessment Data

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Abstract

The viability machinery of the obstruction calculus has been developed and validated on deliberately small exact instances. This paper applies it to a real assessment record — the Northern cod stock (NAFO 2J3KL), read from the published assessment input series, the 2025 reassessment series, the official removals tables, the long-duration catch reconstruction, the research-vessel survey index, and an independently monitored forage-fish covariate — and certifies, in exact rational arithmetic, the obstruction structure that the historical collapse exhibited. The certified object is a *harvest-free multiplier bracket*: for each annual step, the interval’s per-capita non-fishing growth factor lies in $[\rho, \max\{\rho + r, \rho/(1 - r)\}]$, where ρ is the realized net multiplier and r the removals fraction, under either timing convention for the removals accounting. Both collapse steps certify as harvest-free contractions under every convention (upper bounds 0.754 and 0.372); the reference window’s worst step and the two prelude steps do not certify (brackets straddle unity) — a typology, not a blanket verdict. Under the post-moratorium removals rows (1,314 and 413 t, at most 1.4% of the stock), the brackets certify continued contraction, and the stock fell by a factor of $\frac{10105}{968} \approx 10.4$ in two years to its series minimum. The 276 kt auxiliary floor separates the reference window from the collapse era without exception, while the 884.6 kt reference point — the window’s own mean — has three of the seven window readings below it in sample and therefore separates nothing. The collapse is present in the raw survey index (a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further than the assessment series over their common window); the forage-fish covariate collapses in the same annual step in which the mortality diagnostic crosses $M = 1$, two steps before the floor breach; and both assessment vintages re-cross the floor at the 2015 reading, with outcome years at most $\frac{44000}{88460} \approx 49.7\%$ of the reference point. Every quantity is regenerated by an accompanying verification script in exact rational arithmetic.

Keywords: viability; obstruction; regime shift; Northern cod; exact computation; monitoring design; limit reference points

1 Introduction

The viability and obstruction machinery of the calculus — survivable sets, robust multipliers, breach-time certificates, and probe design — is developed in Abaee (2026a) and validated on deliberately small exact instances. This paper applies it to the Northern cod stock (NAFO divisions 2J3KL) and certifies the obstruction structure of its collapse on the public assessment record. The forecast layer of the same record — the scoring of competing forecast ladders and the 2022–2023 observation suspension as a review-timing event — is certified in a companion paper (Abaee, 2026c, Section 3.10); the present paper certifies the viability layer. The two are companions by construction: the forecast ladder prices what can be predicted, this paper prices what could have been kept safe.

The claims below are of three kinds, and the label travels with the claim. *Arithmetic claims* are exact rational identities or inequalities on the certified rows; they are regenerated by the verification script and cannot drift. *Provenance claims* state which file, which vintage, and which column a number comes from, and whether that column is a model input or a model output. *Scope claims* are conditional readings — stress bounds, sustained hypotheticals, design critiques — and are labelled as such where they appear. No claim of the first kind rests on an interpretation; no claim of the third kind is stated as one of the first.

2 Data and provenance

Seven locked public files are used, each cross-verified against its primary source at archival time (Abaee, 2026d): (i) the NCAM 2016 assessment input series (DFO, 2016, Table A2): spawning stock biomass in kilotonnes, 1983–2015, cross-checked 33 of 33 years against the primary report — the *input vintage* of the biomass record; (ii) the xteNCAM 2025 reassessment SSB series (Regular et al., 2025, Table 17), 1954–2024, with 95% bands and SSB-over- B_{lim} ratios — the *reassessment vintage*; (iii) the official removals table (Regular et al., 2025, Table 1), 1954–2023; (iv) the catch reconstruction (Schijns et al., 2021, Table 1); (v) the fall research-vessel survey abundance index (Schijns et al., 2021, Table 3, citing DFO 2021b) — a

Table 1: The record spine, 1983–1995. SSB in kt (assessment input vintage, Table A2); removals in t (official table; the reconstruction gives identical rows for 1992–1995); survey index in the research-vessel file’s index units; covariate in kt, observed years only; M is the assessment’s mean natural-mortality diagnostic (model-estimated). Computed entries are exact rational arithmetic on the file rows; displayed values are rounded.

Year	SSB (kt)	Removals (t)	Survey	Covariate	M
1983	841.08	232,345	2,088,958	—	0.391
1984	863.23	232,471	2,198,605	—	0.377
1985	902.94	231,293	1,288,360	3426	0.349
1986	836.00	266,713	2,502,702	3697	0.277
1987	940.75	239,924	1,020,462	2576	0.494
1988	886.40	268,677	1,223,314	4285	0.335
1989	921.66	253,990	2,127,417	3712	0.289
1990	861.92	219,452	1,627,647	5783	0.403
1991	734.51	172,012	1,117,670	138	1.002
1992	381.95	40,956	239,740	138	2.214
1993	101.05	11,392	90,709	—	2.575
1994	30.55	1,314	21,797	—	2.331
1995	9.68	413	43,240	—	0.288

research-vessel input to the assessments, not an output of them; (vi) the RAM Legacy database per-year extract for this stock (free, 2024 release), 1850–2021 — the *database vintage*, extending the series to 2021; and (vii) the capelin acoustic biomass series (observed years only, no interpolation).

Two thresholds appear throughout, and their provenance is stated once here. The *auxiliary floor* $B_{\text{aux}} = 276$ kt is the reassessment vintage’s B_{lim} at the file’s rounding; the published ratio column is rounded to two decimals, and the values it implies for B_{lim} range from 250 to 289 kt across the file’s years. Every floor comparison below is insensitive across that range (the compared quantities clear it by large factors; Remark 9), so a single representative value is fixed. The *reference point* $B_{\text{ref}} = 884.6$ kt is the input vintage era’s limit reference point, whose documented construction is the 1983–1989 mean of the assessment input series (DFO, 2016); the exact mean is $\frac{44229}{50} = 884.58$ kt (Proposition 1), and 884.6 is its published rounding.

Two discipline rules are respected throughout. First, the mortality and fishing-mortality columns of the assessment input file are diagnostics only (dynamics are never built on them). Second, every annual multiplier certified below is a *realized net* quantity — it conflates the mortality regime with all removals. Section 5 introduces the one decomposition this paper performs on it, which is an accounting bracket, not a biological model.

3 The record spine

Table 1 tabulates the certified rows of the collapse era and its surroundings: assessment input SSB (input vintage), official removals, the survey index, the covariate (observed years only), and the mortality diagnostic. Every proposition below reads its numbers from this table’s files.

4 The reference window and the collapse window

Proposition 1 (seven readings, six transitions; the windows’ worst multipliers). *From the Table A2 series: the reference window 1983–1989 contains seven read-*

ings (841.08, 863.23, 902.94, 836.00, 940.75, 886.40, 921.66) *kt and six annual transitions; their minimum is the 1985 transition* $\rho_g = 836.00/902.94 = \frac{41800}{45147} \approx 0.9259 < 1$, *and their exact mean is* $\frac{44229}{50} = 884.58$ *kt. The collapse window 1991–1993 contains two transitions,* $\frac{38195}{73451} \approx 0.520$ *(1992) and* $\frac{2021}{7639} \approx 0.265$ *(1993), each with numerator strictly below denominator. The connecting 1990 reading is* 861.92 *kt, with transitions* $\frac{86192}{92166} \approx 0.935$ *(1990) and* $\frac{73451}{86192} \approx 0.852$ *(1991).*

Proof. Each multiplier is the exact ratio of consecutive Table A2 SSB entries, computed in the script as a rational of the printed values; the minimum of the six reference-window ratios and the two collapse-window ratios are the displayed fractions by direct comparison. Each displayed fraction is below 1 because its numerator is smaller than its denominator ($41800 < 45147$, $38195 < 73451$, $2021 < 7639$). The mean is the exact rational sum of the seven readings over seven. $\square$

5 Harvest-free multiplier brackets

The net multiplier ρ folds removals into the per-capita rate, so a sub-unitary ρ alone does not certify a non-fishing contraction: with a large enough harvest, even a growing stock declines net of catch. This section isolates the harvest term by an accounting bracket that is exact and robust to the one ambiguity the rows leave open — where in the interval the removals acted.

Lemma 2 (harvest-free multiplier bracket). *Fix a certified interval with readings $B_t, B_{t+1} > 0$ and a removals row $C \geq 0$ assigned to the interval; write $\rho = B_{t+1}/B_t$ and $r = C/B_t$. Suppose the interval’s accounting has one of the two extreme forms, with $g_t > 0$ the interval’s per-capita non-fishing growth factor: $B_{t+1} = g_t B_t - C$ (removals deducted after growth), or $B_{t+1} = g_t(B_t - C)$ (removals deducted before growth). Then respectively $g_t = \rho + r$ or $g_t = \rho/(1 - r)$; in either case*

$$\rho \leq g_t \leq \max\left\{\rho + r, \frac{\rho}{1 - r}\right\},$$

and the upper bound is strictly below 1 if and only if both $\rho + r < 1$ and $\rho/(1 - r) < 1$.

Proof. First form: $g_t = (B_{t+1} + C)/B_t = \rho + r$. Second form: $g_t = B_{t+1}/(B_t - C) = \rho/(1 - r)$, well-defined since $C < B_t$ on every certified interval. Both expressions are $\geq \rho$ (as $r \geq 0$, $1 - r \leq 1$); the disjunction of the two forms gives the displayed maximum, and the final equivalence is direct algebra ($\rho/(1 - r) < 1 \iff \rho < 1 - r$). $\square$

The lemma’s content is accounting, not biology: g_t still conflates natural mortality with recruitment (the regime), but no placement of the *realized* removals within the interval’s accounting — early or late — can leave the non-fishing account non-contracting when the bracket’s upper bound is sub-unitary. The rows do not date removals within the year, so both alignments of the two adjacent rows (C_t, C_{t+1}) are carried; a verdict is certified only when it holds under both conventions and both alignments.

Proposition 3 (the collapse steps are harvest-free contractions; the reference and prelude steps are not certifiable). *Applying Lemma 2 with both alignments of the*

Table 2: Harvest-free multiplier brackets (Lemma 2), both alignments of the removals rows. ρ exact to four decimals; upper bound = $\max\{\rho+r, \rho/(1-r)\}$ in exact rational arithmetic, displayed rounded.

Step	C (t)	ρ	Upper bound	Verdict
1985→86	266,713	0.9259	1.3140	straddles
1989→90	219,452	0.9352	1.2274	straddles
1990→91	172,012	0.8522	1.0646	straddles
1991→92	172,012	0.5200	0.7542	certified
1991→92	40,956	0.5200	0.5758	certified
1992→93	40,956	0.2646	0.3718	certified
1992→93	11,392	0.2646	0.2944	certified
1993→94	1,314	0.3023	0.3153	certified
1994→95	413	0.3169	0.3304	certified

removals rows (Table 2): both collapse-window steps certify harvest-free contraction, the 1992 step under an upper bound of at most 0.7542 and the 1993 step under an upper bound of at most 0.3718, each computed in exact rational arithmetic. The reference window’s worst step (1985–1986, upper bound 1.3140) and the two prelude steps (1989–1990, upper bound 1.2274; 1990–1991, upper bound 1.0646) have brackets that straddle unity: for those steps the certified rows do not decide between a fishing-dominated and a non-fishing-dominated contraction.

Proof. Each row of Table 2 evaluates $\max\{\rho+r, \rho/(1-r)\}$ as an exact rational for the displayed (ρ, C) pair; the collapse-window upper bounds (at most 0.7542) are compared against 1 by numerator–denominator comparison in the script. The prelude and reference rows exceed 1 likewise. $\square$

6 The moratorium obstruction and the post-moratorium experiment

Proposition 4 (the collapse proceeds with the fishery closing; the loss exceeds the removals by certified factors). *Across 1991–1992 the stock lost $734.51 - 381.95 = 352.56$ kt while removals were 40,956 t: the loss exceeds the removals by a certified factor of $\frac{352560}{40956} \approx 8.61$. Across 1992–1993 the loss was $381.95 - 101.05 = 280.90$ kt while removals fell to 11,392 t: a certified factor of $\frac{280900}{11392} \approx 24.66$; removals were 4.06% of the loss. By Lemma 2 the two steps are harvest-free contractions under every timing convention (Proposition 3): the decline over the collapse window is not explainable by the realized removals under any accounting placement of them.*

Proof. The losses are differences of certified readings (exact decimals); the factors are their exact ratios against the removals rows, computed in the script as rationals ($4.06\% = 11392/280900$). The harvest-free attribution is Proposition 3’s certified rows for 1991–1992 and 1992–1993. The statement is a witness about the realized window, not a counterfactual claim about other catch paths. $\square$

Proposition 5 (the reconstruction and the moratorium years). *The reconstruction’s rows for 1992–1995 are 40,956, 11,392, 1,314, 413 t — identical to the official removals table’s rows for the same years (a cross-file equal-*

ity) — and its 2015 row (4,436 t) agrees with the official table and with the assessment report, as recorded at archival time. The moratorium-era removals are at least 176 and at most 651 times below the 1983–1989 removals rows (all above 130,000 t in the same file): a separation of between two and three orders of magnitude ($\frac{231293}{1314} \approx 176.0$ to $\frac{268677}{413} \approx 650.5$).

Proof. The four 1992–1995 rows are read from both removals files and their equalities are exact rational equalities in the script; the 2015 equality is $4436 = 4436$. The scale comparison is the exact minimum and maximum of the four ratios of the 1983–1989 reconstruction rows against the two moratorium rows. $\square$

Proposition 6 (the post-moratorium experiment). *From the 1993 reading 101.05 kt, the certified removals rows are 1,314 t (1994) and 413 t (1995): at most 1.4% of the interval’s start reading in both steps ($r = 1314/101050 \approx 0.0130$ and $413/30550 \approx 0.0135$). Lemma 2 certifies harvest-free contraction in both steps under both conventions (upper bounds 0.3153 and 0.3304): with the fishery effectively closed, the stock fell $101.05 \rightarrow 30.55 \rightarrow 9.68$ kt, a certified factor of $\frac{10105}{968} \approx 10.44$ in two years, and 9.68 kt is the minimum of the entire 1983–2015 Table A2 series.*

Proof. The removals fractions and the bracket upper bounds are exact rationals in the script, compared against 1 by numerator–denominator comparison. The two-year factor is the exact ratio of the 1993 and 1995 readings; the minimum is the exact minimum of the 33 certified readings. $\square$

7 Breach certificates and the cross-vintage check

Proposition 7 (breach certificates on the two thresholds). *With $B_{\text{aux}} = 276$ kt and $B_{\text{ref}} = 884.6$ kt: (i) conditional stress bound: from the end-1991 reading 734.51 kt, one step at the collapse window’s worst multiplier already lands below B_{aux} , since $\frac{2021}{7639} < \frac{27600}{73451}$; this is a hypothetical sustained-rate bound, and the realized breach took two steps. (ii) first below-floor reading: the 1992 reading is above the floor ($381.95 \geq 276$) and the 1993 reading is below it ($101.05 < 276$): the realized first breach of B_{aux} is the 1993 reading, and the input vintage then stays below the floor through 2014 (250.12 kt), re-crossing at 2015 (Proposition 14). (iii) reference point: $734.51 < 884.6$: the 1991 reading is already below B_{ref} , as is the 1990 reading before it ($861.92 < 884.6$).*

Proof. (i) is the exact cross-multiplication $2021 \cdot 73451 = 148,444,471 < 210,836,400 = 7639 \cdot 27600$ in the script. (ii) is the pair of direct rational comparisons on the Table A2 rows, stated so that the intermediate year is visible: the breach is *first* at 1993 only because 1992 is certified above the floor. (iii) is $734.51 < 884.6$ and $861.92 < 884.6$. No step of the proof uses floating point. $\square$

Remark 8 (the breach frontier). On the realized path from the 1989 reading — $921.66 \rightarrow 861.92 \rightarrow 734.51 \rightarrow 381.95 \rightarrow 101.05 \rightarrow 30.55 \rightarrow 9.68$ kt — the first-below reading is a function of the threshold L , certified on intervals: $L \in (861.92, 921.66]$ first falls below at the

1990 reading; (734.51, 861.92] at 1991; (381.95, 734.51] at 1992; (101.05, 381.95] at 1993; (30.55, 101.05] at 1994; (9.68, 30.55] at 1995; and $L \leq 9.68$ is never breached within 1983–2015 (the series minimum is 9.68). The realized threshold $B_{\text{aux}} = 276$ sits in the 1993 interval (Proposition 7(ii)); the reference point $B_{\text{ref}} = 884.6$ sits in the 1990 interval, after breaching repeatedly within its own construction window (Proposition 12).

Remark 9 (the cross-vintage check). The reassessment vintage reads the same three years as 600, 352, 79 kt, with floor ratios 2.18, 1.28, 0.29. The two vintages agree on the sign pattern against the floor — above, above, below — and disagree on levels by certified factors $\frac{73451}{60000} \approx 1.22$, $\frac{38195}{35200} \approx 1.09$, and $\frac{10105}{7900} = \frac{2021}{1580} \approx 1.28$: the reassessment’s 1993 level is 21.8% below the input vintage’s, and its 0.29 ratio is its own level (79 kt) over its own B_{lim} -scale denominator, not the input vintage’s 101.05. Because the ratio column is rounded to two decimals, the implied B_{lim} ranges over 250–289 kt across the file; the sign pattern would flip only if the true threshold were at or below 101.05 kt or above 352 kt, far outside that range, so every floor comparison here is insensitive to the rounding. The cross-file statement certified is therefore agreement on the breach sign pattern, not equality of levels — and the level disagreement itself ($\leq$ a factor of 1.28 between vintages six years apart) is part of the certified record.

8 The instruments

Proposition 10 (the collapse is present in the raw survey index). *The fall research-vessel survey index reads 2,127,417 (1989), 1,117,670 (1991), 239,740 (1992), 90,709 (1993), and 21,797 (1994): a certified factor of $\frac{21797}{2127417} \approx 0.0102$ over 1989–1994, and $\frac{239740}{1117670} \approx 0.214$ across 1991–1992. The index is a research-vessel input to the assessments, not an output of them: the collapse is visible in the raw instrument record, before any model processing. Over the common window 1989–1993 the index falls by $\frac{90709}{2127417} \approx 0.0426$ while the assessment series falls by $\frac{10105}{92166} \approx 0.1096$: the instrument falls 2.57 times further than the model output it feeds.*

Proof. All five values are rows of the locked survey file; the three factors are their exact ratios, and 2.57 is the exact ratio $\frac{10105 \cdot 2127417}{92166 \cdot 90709}$ of the two common-window factors, evaluated in the script. The provenance statement is a property of the file (survey rows are assessment inputs, recorded at archival time). The index-to-biomass mapping assumes constant catchability and coverage; the divergence factor is certified as a difference between the two series’ rates, not attributed. $\square$

Proposition 11 (the covariate crossing is contemporaneous with the mortality crossing). *In the observed capelin series, the acoustic estimate falls from 5783 kt (1990) to 138 kt (1991) — a certified factor of $\frac{138}{5783} \approx 0.024$ — and reads 138 kt again in 1992. The Table A2 mean natural-mortality diagnostic crosses $M = 1$ across the same annual step (0.403 in 1990 to 1.002 in 1991; diagnostic column, model-estimated). At the data’s annual resolution the two crossings coincide; the covariate collapse precedes the certified floor breach (the 1993 reading) by two annual steps*

and the assessment series’ steepest realized step (1991–1992) by one.

Proof. Both covariate readings are observed rows of the locked capelin file (no interpolation, per the archival discipline); the factor is their exact ratio. The mortality crossing is the diagnostic column’s consecutive values straddling 1 across the same pair of years as the covariate collapse — the rows display the coincidence directly, and no sub-annual timing is claimed. Because the diagnostic is model-estimated given assumed catch, its crossing corroborates the regime reading but is not an independent certification of it. $\square$

9 The thresholds as design objects

Proposition 12 (the auxiliary floor separates the eras; the reference point does not). *Against $B_{\text{aux}} = 276$ kt: all seven reference-window readings lie above the floor (minimum 836.00), and the 1993–1995 readings (101.05, 30.55, 9.68) lie below it, with 1990–1992 above: the auxiliary floor separates the reference window from the collapse era without exception. Against $B_{\text{ref}} = 884.6$ kt: three of the seven reference-window readings (841.08, 863.23, 836.00) already lie below, and within 1983–1995 the below-set is $\{1983, 1984, 1986, 1990, \dots, 1995\}$ — nine of thirteen readings: the reference point does not separate the eras. A threshold fixed at a window’s own mean places roughly half that window’s readings below it by construction and retains no headroom above it (the window’s maximum is 6.4% above; its minimum 5.5% below, outward-rounded).*

Proof. All comparisons are direct rational comparisons of the certified readings against 276 and $\frac{8846}{10}$ in the script; the below-sets are computed by exact membership. The mean-rule statement is Proposition 1’s exact mean identity: a threshold at the mean of a set of readings lies between the set’s minimum and maximum, so the set straddles it. $\square$

Proposition 13 (conditional: sustained worst-case contraction breaches the floor in sixteen steps). *Under the reference window’s worst realized multiplier $\rho_g = \frac{41800}{45147}$ sustained from the 1989 reading 921.66 kt — a hypothetical rate, held fixed — the stock breaches $B_{\text{aux}} = 276$ kt within sixteen steps: $\rho_g^{16} \leq \frac{27600}{92166} < \rho_g^{15}$. The count is a ceiling convention (15.66 steps rounds up); the statement is conditional on the sustained hypothetical rate and carries no harvest-free reading: the window’s own bracket straddles unity (Proposition 3), so the sustained-rate contraction is not attributable to the non-fishing account by the certified method.*

Proof. The sixteenth and fifteenth integer powers of $\frac{41800}{45147}$ are compared against $\frac{27600}{92166}$ by exact cross-multiplication in the script. The ceiling convention is stated because the exact threshold step lies strictly between the fifteenth and sixteenth powers. $\square$

10 The outcome years

Proposition 14 (both vintages re-cross the floor at the 2015 reading; outcome years stay below half the reference

Table 3: The certified quantities and their files. Computed in exact rational arithmetic on the locked rows; displayed values rounded.

Quantity	Value	Source file
Ref-window worst mult. (1986)	41800/45147	Table A2
Collapse multipliers (1992, 1993)	0.520, 0.265	Table A2
Collapse-step bracket bounds	0.7542, 0.3718	A2 + T1
Post-moratorium 2-yr factor	10105/968 $\approx$ 10.44	Table A2
Removals 1992, 1993	40,956, 11,392	Table 1 (2025)
Removals 1994, 1995	1,314, 413	T1 = reconstr.
Survey index 1989, 1994	2,127,417, 21,797	Survey T3
Covariate 1990, 1991	5783, 138	Capelin file
First below-floor reading	1993 (101.05 < 276)	Table A2
Cross-vintage 1993 levels	101.05 vs 79 kt	A2 + T17
Outcome years 2016–2021	340–440 kt	DB extract

point). *Each vintage’s own series re-crosses $B_{\text{aux}} = 276$ kt at the 2015 reading: the input vintage goes $250.12 \rightarrow 298.65$ kt (2014–2015), the database vintage $238 \rightarrow 277$ kt. The database vintage then reads 340, 433, 394, 419, 440, 411 kt for 2016–2021 (six outcome years, all above the floor), with maximum 440 kt at $\frac{44000}{88460} \approx 49.7\%$ of B_{ref} . The recovery’s certified removals rows are 4.4 to 12.9 kt per year (2015–2021, official table; maximum 12,881 t in 2017) — between 2.3% and 3.0% of the same vintage’s stock reading per year.*

Proof. The crossing pairs are direct rational comparisons in the two locked files; the six outcome values are the database rows (exact integers in kt); the maximum comparison is the reduced ratio $\frac{44000}{88460}$. The removals range is the exact minimum and maximum of the seven official rows for 2015–2021. $\square$

Remark 15 (the monitoring consequence). Read together: the certified obstruction is a property of the dynamics under the realized catch (Propositions 3 and 4), the raw instrument record shows the collapse before the model output does (Proposition 10), the covariate crossing is coincident with the mortality crossing and two steps ahead of the floor breach (Proposition 11), and the mean-rule reference point separates nothing (Proposition 12). The assessment machinery itself was reconstructed around a monitoring indicator after 2016 (a three-year review cycle with an interim survey indicator, the indicator value available by early January) — the same review-timing reading the companion paper certifies for the 2022–2023 observation suspension (Abaee, 2026c, Section 3.10).

11 Delimitations

Every arithmetic claim is a certificate about the realized rows of the certified files — not a forecast, not a stochastic model, and not a counterfactual catch analysis. The bracket of Lemma 2 is an accounting decomposition: it removes the realized removals from the per-capita rate under two extreme timing conventions, and its harvest-free factor still conflates natural mortality with recruitment — the certified statement is that no placement of the realized removals explains the decline, not that no catch path could have. The biomass series are assessment outputs (model-based), treated as the data of record with their published uncertainty untouched; the exact arithmetic is exact on the published point estimates, not a claim about the latent stock. The mortality diagnostic is model-estimated given

assumed catch, so the mortality-crossing statement corroborates the regime reading without independently certifying it; the survey index carries its own sampling error and catchability assumptions, untouched here, and the instrument–output divergence factor is certified as a difference of rates, not attributed to a cause. The covariate series is observed-years only; the contemporaneity statement is about certified rows at annual resolution. Single stock; two vintages of one threshold and one vintage of the reference point; no averaging beyond the reference point’s own construction.

12 Verification methods

Every number, fraction, inequality, power, ratio, bracket, and temporal claim in this paper is regenerated by the accompanying script in exact integer and rational arithmetic, standard library only, directly from the locked files: the seven reference-window readings with their six multipliers, exact mean ($\frac{44229}{50}$), and minimum; the collapse-window and connecting multipliers; the harvest-free brackets of Table 2 under both alignments, each upper bound compared against unity by numerator–denominator comparison; the loss factors $\frac{352560}{40956}$ and $\frac{280900}{11392}$ and the removals share 11392/280900; the cross-file equality of the 1992–1995 removals rows and the 2015 row; the moratorium scale ratios ($\frac{231293}{1314}$ to $\frac{268677}{413}$); the post-moratorium brackets and the two-year factor $\frac{10105}{968}$; the breach comparisons by integer cross-multiplication ($2021 \cdot 73451 < 7639 \cdot 27600$) and the first-below-floor pair with the 1992 intermediate; the cross-vintage sign pattern, level factors, and implied-threshold range; the survey factors and the divergence ratio; the covariate and mortality rows; the threshold-discrimination below-sets; the sustained sixteenth-step comparison; the two-vintage re-crossings, outcome values, and removals range; and every headline number of this text asserted present verbatim.

13 Conclusion

Applied to the public record of the Northern cod, the calculus certifies the following. The collapse window’s two annual steps are harvest-free contractions under every timing convention the rows admit (upper bounds 0.754 and 0.372): no placement of the realized removals — which the year of the floor breach puts at 4.06% of the stock’s loss — explains the decline, and under the moratorium’s own removals rows the non-fishing account drove a further certified factor-of-10.4 contraction to the series minimum. The reference window’s worst step and the two prelude steps do not admit that reading: their brackets straddle unity, so the certified record supports a typology (two harvest-free collapse steps inside a fishing-era decline), not a blanket verdict about the decade. The 276 kt auxiliary floor separates the eras without exception, while the 884.6 kt reference point — fixed at the window’s own mean — has three of the seven window readings below it in sample and separates nothing: a design property of mean-rule thresholds, certified here on the readings themselves. The collapse is present in the raw survey index at a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further over the common window than the assessment series it

feeds; the covariate crossing coincides with the mortality crossing at annual resolution and leads the floor breach by two steps. Both assessment vintages re-cross the floor at the 2015 reading and the outcome years stay below half the reference point. These are exact statements about published point estimates; their scope is the realized record, and their monitoring content — which instruments moved, when, relative to which threshold — is the applied content the obstruction calculus prices.

References

- DFO (2016). Stock assessment of Northern cod (NAFO 2J3KL). *Can. Sci. Advis. Sec. Sci. Advis. Rep.* 2016/026.
- Regular, P. et al. (2025). Northern cod (NAFO 2J3KL) reassessment. *DFO Can. Sci. Advis. Sec. Res. Doc.* 2025/048.
- Schijns, R. et al. (2021). Reconstructing three centuries of fish and fisheries catch in the Northwest Atlantic. *ICES Journal of Marine Science* 78, 2675–2684.
- Abae, A. (2026a). An obstruction calculus for viability under incomplete observation. Submitted manuscript.
- Abae, A. (2026c). The Northern cod forecast ladder under a frozen specification. Preprint.
- Abae, A. (2026d). Calibration data record, Northern cod (NAFO 2J3KL): sources, cross-checks, and locked extracts. Preprint.

Declarations

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None.

Competing interests

None.

Data availability

All inputs are the archived public files of the calibration record (wave-e-cod extracts and the record’s CSV files), with primary sources cited at archival time.

Code availability

The verification script `applied_regime_viability_v4_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates every fraction, inequality, bracket, power, ratio, and temporal claim verbatim from the locked files.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.